Lab#04
Implementation of various operators in java

Objective

To get familiarity with the various operators in java.

Lab Goals

To provide working insight into the usage of various operators in java programming:

a. Demonstrate the implementation of various operators.

b. Explore Arithmetic, Assignment, Relation, Logical, Ternary and Bitwise operators in Java.

c. Explore Auto-increment and Auto-decrement Operators

Introduction to operators

In Java, operators are symbols used to perform operations on variables and values. They are categorized into different types based on the operations they perform, such as arithmetic, relational, logical, bitwise, assignment, and more.

Basic Arithmetic Operators

These operators are used to perform basic arithmetic operations like addition, subtraction, multiplication, division, and modulus.

- **+ (Addition)**
- **- (Subtraction)**
- *** (Multiplication)**
- **/ (Division)**
- **% (Modulus)**

Example

```
public class Main {
   public static void main(String[] args) {

      // Arithmetic Operators
      int a = 10;
      int b = 5;
      int sum = a + b;        // Addition
      int difference = a - b; // Subtraction
      int product = a * b;    // Multiplication
      int quotient = a / b;   // Division
      int remainder = a % b;  // Modulus
```

Exercise

A. Create a Java application for a basic calculator with the following capabilities: Addition and Subtraction, Division, Exponentiation, Multiplication, also use bitwise operator and Ternary operator to find the maximum, minimum value.

Program:

```java
import java.util.Scanner;
public class Lab4 {

    Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter first number: ");
        int a = sc.nextInt();

        System.out.print(s: "Enter second number: ");
        int b = sc.nextInt();

        // Addition
        System.out.println("Addition: " + (a + b));

        // Subtraction
        System.out.println("Subtraction: " + (a - b));

        // Multiplication
        System.out.println("Multiplication: " + (a * b));
```

```java
        // Division
        System.out.println("Division: " + (a / b));

        // Exponentiation
        System.out.println("Exponentiation: " + (int) Math.pow(a, b));

        // Bitwise AND
        System.out.println("Bitwise AND: " + (a & b));

        // Bitwise OR
        System.out.println("Bitwise OR: " + (a | b));

        // Ternary - Maximum
        int max = (a > b) ? a : b;
        System.out.println("Maximum: " + max);

        // Ternary - Minimum
        int min = (a < b) ? a : b;
        System.out.println("Minimum: " + min);

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 4> java Lab4
Enter first number: 2
Enter second number: 4
Addition: 6
Subtraction: -2
Multiplication: 8
Division: 0
Exponentiation: 16
Bitwise AND: 0
Bitwise OR: 6
Maximum: 4
Minimum: 2
PS C:\Users\Ansharah\Desktop\Java\Lab 4>
```

B. Create a program that demonstrates operator precedence by showing how parentheses can change the result of complex expressions.

Hint:

- Initialize variables with values that will clearly show precedence differences
  int a = 10, b = 5, c = 2, d = 3;
- **Expression 1:** Without parentheses
- **Expression 2:** With parentheses changing the order

Program:

```java
public class Lab4 {

    Run | Debug
    public static void main(String[] args) {

        int a = 10, b = 5, c = 2, d = 3;

        System.out.println("Values: a=" + a + ", b=" + b + ", c=" + c + ", d=" + d);
        System.out.println(x: "========================================");

        // Expression 1: Without parentheses
        int result1 = a + b * c - d;
        System.out.println(x: "Expression 1 (Without parentheses):");
        System.out.println("a + b * c - d = " + a + " + " + b + " * " + c + " - " + d);
        System.out.println("Step: b*c = " + b + "*" + c + " = " + (b * c));
        System.out.println("Step: a + 10 - d = " + a + " + 10 - " + d);
        System.out.println("Result = " + result1);

        System.out.println(x: "----------------------------------------");
```

```java
        // Expression 2: With parentheses changing the order
        int result2 = (a + b) * (c - d);
        System.out.println(x: "Expression 2 (With parentheses):");
        System.out.println("(a + b) * (c - d) = (" + a + "+" + b + ") * (" + c + "-" + d + ")");
        System.out.println("Step: (a+b) = " + (a + b));
        System.out.println("Step: (c-d) = " + (c - d));
        System.out.println("Result = " + result2);

        System.out.println(x: "========================================");
        System.out.println("Without parentheses result : " + result1);
        System.out.println("With parentheses result    : " + result2);
        System.out.println(x: "Results are DIFFERENT — parentheses change the answer!");
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 4> java Lab4
Values: a=10, b=5, c=2, d=3
==========================================
Expression 1 (Without parentheses):
a + b * c - d = 10 + 5 * 2 - 3
Step: b*c = 5*2 = 10
Step: a + 10 - d = 10 + 10 - 3
Result = 17
------------------------------------------
Expression 2 (With parentheses):
(a + b) * (c - d) = (10+5) * (2-3)
Step: (a+b) = 15
Step: (c-d) = -1
Result = -15
==========================================
Without parentheses result : 17
With parentheses result    : -15
Results are DIFFERENT ? parentheses change the answer!
PS C:\Users\Ansharah\Desktop\Java\Lab 4>
```

D. To convert temperature from Celsius to Fahrenheit, write a Java program

Guidelines:
Asking for the temperature in Celsius will prompt the user.
Check the temperature that the user has entered.
Use this formula to convert the temperature from Celsius to Fahrenheit: Degrees Celsius * 9/5) + 32 = Fahrenheit

Print the Fahrenheit temperature conversion.

Program:

```java
import java.util.Scanner;

public class Lab4{

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // Ask user for temperature in Celsius
        System.out.print(s: "Enter temperature in Celsius: ");
        double celsius = sc.nextDouble();

        // Convert Celsius to Fahrenheit using the formula
        double fahrenheit = celsius * 9 / 5 + 32;

        // Print the result
        System.out.println(celsius + " Celsius = " + fahrenheit + " Fahrenheit");

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 4> javac Lab4.java
PS C:\Users\Ansharah\Desktop\Java\Lab 4> java Lab4
Enter temperature in Celsius: 100
100.0 Celsius = 212.0 Fahrenheit
PS C:\Users\Ansharah\Desktop\Java\Lab 4>
```

E. Write a program which shows the implementation of increment and decrement operators.

Program:

```java
public class Lab4 {

    Run | Debug
    public static void main(String[] args) {

        int a = 5;

        System.out.println("Initial value of a = " + a);
        System.out.println(x: "================================");

        // Pre-Increment (++a) — increment first, then use
        System.out.println("Pre-Increment (++a) = " + (++a));   // a becomes 6

        // Post-Increment (a++) — use first, then increment
        System.out.println("Post-Increment (a++) = " + (a++));  // prints 6, then a becomes 7
        System.out.println("After Post-Increment, a = " + a);   // a is now 7

        System.out.println(x: "================================");

        // Pre-Decrement (--a) — decrement first, then use
        System.out.println("Pre-Decrement (--a) = " + (--a));   // a becomes 6
```

```java
        // Post-Decrement (a--) — use first, then decrement
        System.out.println("Post-Decrement (a--) = " + (a--));  // prints 6, then a becomes 5
        System.out.println("After Post-Decrement, a = " + a);   // a is now 5

        System.out.println(x: "================================");
        System.out.println("Final value of a = " + a);
    }
}
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 4> javac Lab4.java
PS C:\Users\Ansharah\Desktop\Java\Lab 4> java Lab4
Initial value of a = 5
================================
Pre-Increment (++a) = 6
Post-Increment (a++) = 6
After Post-Increment, a = 7
================================
Pre-Decrement (--a) = 6
Post-Decrement (a--) = 6
After Post-Decrement, a = 5
================================
Final value of a = 5
PS C:\Users\Ansharah\Desktop\Java\Lab 4>
```

F. Write a program which shows the implementation of Ternary Operator.
Conditions:
- Check even and odd number.
- Check Absolute value.

Program:

```java
import java.util.Scanner;
public class Lab4 {

    Run | Debug
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        // ---- Check Even or Odd ----
        System.out.print(s: "Enter a number to check Even/Odd: ");
        int num = sc.nextInt();

        String result = (num % 2 == 0) ? "Even" : "Odd";
        System.out.println(num + " is " + result);

        System.out.println(x: "================================");

        // ---- Check Absolute Value ----
        System.out.print(s: "Enter a number to find Absolute value: ");
        int n = sc.nextInt();

        int absolute = (n < 0) ? -n : n;
        System.out.println("Absolute value of " + n + " = " + absolute);

        sc.close();
```

Output:

```
PS C:\Users\Ansharah\Desktop\Java\Lab 4> javac Lab4.java
PS C:\Users\Ansharah\Desktop\Java\Lab 4> java Lab4
Enter a number to check Even/Odd:
99
99 is Odd
================================
Enter a number to find Absolute value: 45
Absolute value of 45 = 45
PS C:\Users\Ansharah\Desktop\Java\Lab 4>
```